Cocoa/hot-chocolate beverage mixes and malted drink powders: HMTc Standard

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CORRESPONDENCE

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heavy metals, food safety, dietary exposure

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Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for cocoa/hot-chocolate beverage mixes and malted drink powders in Beverages, evaluated on dry powder as sold, reported on the as-sold article, which is the basis EU 2023/915 and EU 2024/1987 use for the cocoa-powder cells that bind this row (product as placed on the market). 2 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit-only): the strictest applicable sovereign maximum level for this exact beverage form and analyte in the row's declared native basis, with disclosed read-across where the exact cell is unregulated, a substantiated not-material determination where a species is bounded or is a method artifact, and a control where no defensible sovereign number and no supportable near-neighbour exist. Never occurrence-derived. Ready-to-drink and dry-commodity bases are never mixed and no brewing, dilution or reconstitution factor is invented.

2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	control	measured every lot; no human-food maximum

Cadmium (Cd)	Tier 1	500	government maximum level
Total arsenic (tAs)	Tier 2	control	measured every lot; no human-food maximum
Inorganic arsenic (iAs)	Tier 1	control	measured every lot; no human-food maximum
Mercury, total (tHg)	Tier 1	control	measured every lot; no human-food maximum
Methylmercury (MeHg)	Tier 1	n/m	not material in this matrix
Nickel (Ni)	Tier 2	15,000	government maximum level
Tin (Sn)	Tier 2	control	measured every lot; no human-food maximum
Aluminium (Al)	Tier 2	control	measured every lot; no human-food maximum
Total chromium (Cr)	Tier 2	control	measured every lot; no human-food maximum
Hexavalent chromium (Cr-VI)	Tier 2	control	measured every lot; no human-food maximum

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION

Lead (Pb)	control	No credible sovereign sets a lead maximum level for cocoa powder, drinking chocolate or a malted drink powder. This is a real and consequential gap, and it is disclosed rather than filled. EU 2023/915 writes cadmium cells for cocoa powder and for every chocolate cocoa-solids band but no lead cell for any of them; Codex CXS 193, FSANZ Schedule 19, Health Canada and Hong Kong Cap 132V likewise set no lead maximum level for cocoa powder, drinking chocolate or malted drink powder. The read-across candidates were examined and all rejected: the Hong Kong Milk cell (20 ug/kg) is an as-consumed liquid value and applying it to a dry powder that is reconstituted at roughly one part in ten would be a basis conflation in the strict direction; the sugar cells cover a different, refined commodity that supplies little of a mix's lead; the tea and coffee commodity cells are unrelated botanicals. Lead is the analyte of greatest concern in cocoa supply chains, so this control is written with a hard escalation rule rather than a soft one. Lead is a Tier-1 analyte, so a control here means mandatory lot testing with an escalation rule, not an absence of scrutiny; no ceiling is invented from occurrence data.
Cadmium (Cd)	500	government maximum level

Total arsenic (tAs)	control	No credible sovereign sets a total-arsenic maximum level for cocoa powder, drinking chocolate or a malted drink powder. Regulators write beverage arsenic limits on the inorganic species (EU 2023/915 section 3.4.4 fruit juices at 20 ug/kg and section 3.4.1.6 non-alcoholic rice-based drinks at 30 ug/kg; the US FDA apple-juice action level at 10 ug/kg) and write unspeciated arsenic limits only in the packaged-water and public-water-supply schedules, which are Category 8 instruments. No sovereign writes an inorganic-arsenic cell for this commodity either, so both arsenic cells in this row are controls and the escalation rule carries the obligation. Total arsenic is therefore mandatory on every lot as the parent screen and no total-arsenic ceiling is invented.
Inorganic arsenic (iAs)	control	No credible sovereign sets an inorganic-arsenic maximum level for cocoa powder, drinking chocolate or a malted drink powder. Every sovereign inorganic-arsenic cell is scoped either to rice and rice-based products, to a ready-to-drink beverage (EU fruit juices 20 ug/kg, EU non-alcoholic rice-based drinks 30 ug/kg, US FDA apple juice 10 ug/kg), to seafood, or to packaged and public drinking water. None of those is applicable to this dry commodity, and applying a ready-to-drink beverage cell to a dry leaf, bean or powder would be a basis conflation. Inorganic arsenic is a Tier-1 analyte, so it is speciated whenever the total-arsenic screen is elevated.

Mercury, total (tHg)	control	No credible sovereign sets a total-mercury maximum level for cocoa powder, drinking chocolate or a malted drink powder. The sovereign mercury values in this basis family are the packaged--drinking-water quality standards (US 21 CFR 165.110 total mercury at 2 ug/kg; Hong Kong Cap 132V packaged drinking water, written for inorganic mercury at 6 ug/kg) and the fish and seafood methylmercury and total-mercury ceilings, none of which governs this product form. Mercury has no identified enrichment pathway in this supply chain, but the analyte is on the mandatory eight-total panel, so testing is mandatory and no ceiling is invented.
Methylmercury (MeHg)	n/m	Methylmercury has no formation pathway in a dry cocoa or malted beverage powder: there is no marine bioaccumulation step in the supply chain and no anaerobic, sulfate-reducing methylation environment in the product or its process. Methylmercury is a component of total mercury (MeHg <= tHg) and is governed by and bounded by the total-mercury cell; no separate methylmercury ceiling is published.
Nickel (Ni)	15,000	government maximum level

Tin (Sn)	control	No credible sovereign sets a tin maximum level for a dry cocoa or malted beverage powder. The exclusion is written into the instrument itself: EU 2023/915 Annex I section 3.5 scopes its inorganic-tin maximum levels to canned beverages and canned foods and expressly excepts canned dried and canned powdered products, and the Codex, FSANZ and Hong Kong canned-food cells carry the same carve-out. There is therefore no sovereign number for a dry beverage commodity and no supportable read-across, because the canned-liquid cells are written in a ready-to-drink basis. Tin is nonetheless on the mandatory eight-total panel.
Aluminium (Al)	control	No credible sovereign sets an aluminium maximum level for cocoa powder, drinking chocolate or a malted drink powder, and no near-neighbour read-across is supportable: the only sovereign aluminium values are packaged--drinking-water quality standards written in a litre-of-water basis (Category 8) and the FSANZ infant-formula limits (Category 1). Applying a water-basis value to a dry commodity would be a basis conflation of roughly three orders of magnitude. Aluminium is a live cell for this row: cocoa is grown on acidic tropical soils where plant-available aluminium is elevated, and alkalised (Dutch-process) cocoa carries a processing-aid route as well. Both are investigated on an elevated result.

Total chromium (Cr)	control	No credible sovereign sets a total-chromium maximum level for cocoa powder, drinking chocolate or a malted drink powder. The only sovereign chromium values reachable from this basis family are the packaged-drinking-water quality standards (US 21 CFR 165.110 at 100 ug/kg, Hong Kong Cap 132V packaged drinking water at 50 ug/kg), which govern water sold as water and are Category 8 instruments, and the public-water-system maximum contaminant levels (US EPA, Health Canada), which are water-utility instruments and not food law at all. Total chromium in this form is dominated by benign Cr(III) and is reported as a stainless-steel equipment and process-water leaching screen, not as a health limit; the health-relevant species is addressed in the Cr-VI cell.
Hexavalent chromium (Cr-VI)	control	No credible sovereign sets a hexavalent-chromium limit for any food or beverage, and the in-matrix reduction argument that supports a not-material determination in an acidic beverage is not available here: this row's certified article is a DRY powder, and alkalised cocoa is mildly ALKALINE rather than acidic, which is the condition under which Cr(VI) is most stable rather than least; the acidic-beverage reduction argument is therefore not merely unavailable but pointing the other way. Because Cr(VI) cannot be assumed to be reduced to Cr(III) in this matrix, the fail-safe disposition is a control rather than a not-material determination, and no ceiling is invented.

4 | Cite this standard

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STATUS AND LICENCE

COMPETING INTERESTS

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/cocoa--beverage-mixes. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/-cocoa-beverage-mixes.